

Occupational projections are the prediction of employment by occupation. The State of Florida currently uses the Standard Occupational Classification system to determine how to group jobs by occupation and follows the BLS methodology to predict the employment of these groups. This method involves “statistical and econometric models combined with subjective analysis.”¹ The methodology makes use of several categories of data points. The first step in the process is to collect labor force data by several characteristics including age, sex, race, and ethnicity.² Aggregate Economy data points include GDP, total employment, and major demand categories.³ The Final Demand category of data points includes sales to consumers, businesses, government, and foreigners.⁴ The input-output model uses industry input and outputs to project commodity demand.⁵ “The regression model utilizes industry output, industry wage rate relative to industry output prices, and time. From these estimates of hours, projected wage and salary employment by industry is derived.”⁶ These steps lead to the data being inputted into models to predict occupational employment. The State of Florida relies on BLS for the majority of the inputs and produces only industry and occupational employment projections based on the inputs.

The current occupational projections process used by the State of Florida is tied to the mandatory methodology set by the Bureau of Labor Statistics. This process does not involve machine learning. Because machine learning is a method focused on accuracy and producing results as opposed to explaining results like current econometric and statistical methods, implementing machine learning into this process may result in more accurate projections. The implementation of machine learning in what is typically a statistical and econometrics space is

¹ <https://www.bls.gov/opub/hom/emp/calculation.htm#general-assumptions>

² <https://www.bls.gov/emp/documentation/projections-methods.htm#labor>

³ <https://www.bls.gov/emp/documentation/projections-methods.htm#aggregate>

⁴ https://www.bls.gov/emp/documentation/projections-methods.htm#final_demand

⁵ https://www.bls.gov/emp/documentation/projections-methods.htm#industry_output

⁶ https://www.bls.gov/emp/documentation/projections-methods.htm#industry_employment

significant within the field of data science because it is evidence of the effectiveness and versatility of machine learning. In the econometrics space, the findings of this paper may be significant because of the subjective analysis involved in the current process. Finding an automated and more systematic way of generating these employment projections would likely not only increase the efficiency of the agency where these projections are produced but also the accuracy of the projections themselves. However, for this to be true, the correct models and methods need to be selected.

Kyungsu Kim found that Random Forest was the most accurate in predicting unemployment data.⁷ This lead influences the methods for this paper as a model that can accurately predict unemployment data is likely to have success with employment data as well. J. G. Sukumar, et al. also found that Random Forest was the most accurate model for their data as well. This finding by Sukumar, et al. was more significant to this paper because they used more similar data than Kim. Sukumar, et al. used occupational data and similar inputs to predict wages. While this paper is not predicting wages, it is using occupational data and features similar to the others used as inputs in Sukumar, et al.

This paper is meant to test the accuracy of machine learning models, specifically XGBoost, in predicting occupation employment compared to the current BLS method used by the State of Florida. It may also be helpful to find out which features are contributing the most to the accuracy of the prediction. The features used in the data set include historical employment, mean and median wages, entry and experiences wages, and BLS' percentile rank of the given

⁷ <https://doi.org/10.48550/arXiv.2505.01933>

occupation in each skill category. The skill categories theorized to have the most predictive power are “Computers and Information Technology” and “Critical and Analytical Thinking.”

This paper will illustrate the accuracy of models such as XGBoost, Linear Regression, Lasso Regression, and Ridge Regression. After a preliminary run of the data on the Random Forest model, it was decided that the model was too costly to run due to its run time.

Bibliography

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